

Public CAD Is Not Operational Ground Truth

Regime Change, Hurricane Ida, and the Measurement of Police Responsiveness

25 August 2026

Abstract

Public computer-aided dispatch (CAD) files expose administrative fields that are often converted into operational clocks. This paper measures a more limited object: whether the released `TimeDispatch` and `TimeArrive` fields are present and valid on each non-officer-initiated record, and how their four joint field states change across windows. These are public-record completeness and configuration measures, not response times or proof of physical dispatch and arrival. In New Orleans, the arrival-present/dispatch-absent state first appears on 28 July 2021, with a coordinated change in the officer-initiated stream. Five weeks later, Ida produces a two-day reconfiguration with an unstandardized full-count maximum contrast of 0.5320, rank 1/152 including Ida against 151 post-hoc post-change references. The standardized calculation covers 86.1 percent of event records and 77.0 percent of baseline records and fails the references' 0.90 eligibility rule, making the full-count result essential. In a conditional window-wise multinomial bootstrap that does not model temporal dependence, Ida ranks first in all 4,000 draws. Public documents clarify field labels and exclude two technology explanations without identifying the July cause. Public CAD therefore supports reproducible administrative-field measures, not automatic measures of physical response, capacity, service quality, or domestic-violence incidence.

Keywords: calls for service; police responsiveness; administrative data; Hurricane Ida; measurement error; partial identification

JEL codes: C18, C55, H12, K42

1 Introduction

CAD systems ordinarily record milestones in the handling of a call: creation, dispatch, arrival, and closure. Response-time research often subtracts one released timestamp from another. This paper first asks a simpler measurement question. For every released record, are the public dispatch and arrival fields populated and usable, and does their joint pattern remain stable over time? The resulting shares describe the completeness and configuration of the released administrative record; they are not themselves durations, response outcomes, or physical-event indicators.

Public CAD files nevertheless appear to record a direct sequence: a call is created, an officer is dispatched, and an arrival is entered. That sequence becomes a performance measure only if the released fields retain stable meanings, coverage, and lineage. A missing dispatch timestamp can reflect an operational event, a workflow exception, later entry, retention, redaction, or export logic. These pathways are observationally equivalent in a public file.

Write D_t^{CAD} for reported demand entering CAD, C_t for effective capacity, and P_t for the priority process. Operations produce a service path

$$S_t = g(D_t^{\text{CAD}}, C_t, P_t), \quad (1)$$

while recording, retention, and export R_t^{rec} produce the released record

$$Y_t = h(S_t, R_t^{\text{rec}}). \quad (2)$$

A public timestamp is a feature of Y_t , not proof of its latent counterpart in S_t .

Hurricane Ida makes this distinction visible, but the first empirical result predates the hurricane. In the official 2021 file, a new arrival-without-dispatch state begins on 28 July. The raw audit shows a coordinated change across initiation streams: dispatch fields largely disappear from officer-initiated records while the same missing-dispatch configuration begins appearing among non-officer-initiated records. Ida occurs five weeks into this changed record regime and adds a temporary, unusually large reconfiguration.

The paper contributes four results. First, it binds the late-July break to the official monthly source bytes and shows that it is not an artifact of the aggregate pipeline. Second, it separates a common-support standardized comparison from a genuinely support-independent full-count comparison. Third, it uses public institutional documents to define what is and is not learned about the change: labels are clearer and two technology stories are inconsistent with the chronology, but the cause remains unidentified. Fourth, it translates the findings into a measurement rule for response-time and reported-domestic-violence research: declare the call denominator, public clock, endpoint coverage, and lineage before interpreting field durations as service performance.

Main finding. The joint configuration of released CAD fields changes before Ida and changes again during Ida. The public evidence documents a released-record discontinuity and an event-window field reconfiguration. It does not identify physical dispatch, physical arrival, effective capacity, response quality, domestic-violence incidence, or the mechanism generating the public fields.

2 Related literature

2.1 Response time and first-responder performance

Response delays matter for crime clearance, victim harm, and public-service costs. Faster police arrival can improve clearance and reduce injuries, while congestion and staffing influence first-responder travel and queueing (Blanes i Vidal and Kirchmaier, 2018; DeAngelo et al., 2023; Brent and Beland, 2020; Mourtgos et al., 2024). A broader policing literature stresses that performance is multidimensional and that a single administrative clock may reward a selected stage rather than the public objective (Moore and Braga, 2004; Tiwana et al., 2015; Verlaan and Ruiter, 2023). This paper contributes a prerequisite: before estimating response-time determinants or consequences, establish that the public start and end fields describe a stable measurement object on a declared denominator.

2.2 Administrative records and measurement error

Administrative data are generated by institutions rather than designed solely for research. Recorded outcomes can reflect classification practice, verification, discretion, and incentives (Klinger and Bridges, 1997; Boivin and Cordeau, 2011; Boivin and Ouellet, 2014; Simpson and Orosco, 2021). Misclassification and measurement error can distort both group comparisons and structural interpretation (Kapteyn and Ypma, 2007; Chen et al., 2011; Knox et al., 2020). Government performance systems can also alter the recorded object that is optimized or reported (Courty and Marschke, 1997; Eckhouse, 2022). The distinctive evidence here is a sharp change in the joint field-presence distribution before any duration is analyzed.

2.3 Disasters and domestic violence

Disaster exposure, displacement, psychological stress, partner conflict, and service demand can move together (Anastario et al., 2009; Schumacher et al., 2010; Harville et al., 2011; First et al., 2022). Police composition can affect crime reporting and domestic-violence reporting (Miller and Segal, 2019). Pandemic-era studies show that calls, reports, and underlying victimization need not move one-for-one (Leslie and Wilson, 2020; Bullinger et al., 2021; Miller et al., 2022, 2024). These studies motivate attention to reported domestic-violence calls during emergencies, but they do not validate a particular CAD clock. Ida is used here as a system-level stress test of administrative observability, not as a treatment whose effect on domestic-violence incidence is estimated.

2.4 Partial identification and honest uncertainty

When the mapping from latent events to observed records is incomplete, point identification should not be assumed (Manski, 1989, 1990, 2003; Tamer, 2010; Molinari, 2020). Misclassified outcomes can instead support bounds under transparent restrictions (Horowitz and Manski, 1995; Molinari, 2008). That logic informs two choices here. Public-field configurations are analyzed directly without assigning them latent operational meanings, and downstream reported-DV measures retain 1/0/U classification and sharp bounds when classification or endpoint authority is unresolved.

Relative to these literatures, the paper contributes three findings. First, it documents a date-localized change in the joint presence of public CAD fields rather than assuming a temporally stable administrative measurement regime. Second, it shows that Ida produces an additional extreme but temporary reconfiguration within the later regime. Third, it translates those facts into a measurement requirement: response-time and reported-DV analyses must validate the denominator, public endpoints, endpoint coverage, and stage lineage before interpreting released durations as operational performance.

3 Data and measurement

3.1 Official files and the declared denominator

The analysis uses the official annual New Orleans Calls for Service datasets for 2020–2024 and the official 2025 and 2026 files (City of New Orleans, 2020, 2021, 2022, 2023, 2024, 2025, 2026). The source audit binds each annual dataset identifier to twelve cached monthly files, observed date limits, row counts, and a deterministic SHA-256 bundle digest. The primary denominator is *non-officer-initiated public CAD records*, identified by the published `SelfInitiated` field. Officer-initiated records are a separate measurement stream, not extra observations silently pooled into the main denominator.

Table 1. Raw official-file audit around 28 July 2021

Date	Stream	Records	Dispatch present	Arrival valid	J_{01}
25 Jul.	Non-officer	721	631	532	0
26 Jul.	Non-officer	733	663	565	0
27 Jul.	Non-officer	731	658	585	0
28 Jul.	Non-officer	620	509	502	34
29 Jul.	Non-officer	666	510	532	69
30 Jul.	Non-officer	705	542	559	65
31 Jul.	Non-officer	725	584	572	45
27 Jul.	Officer	479	479	479	0
28 Jul.	Officer	575	230	575	345
29 Jul.	Officer	504	5	504	499
30 Jul.	Officer	548	3	548	545
31 Jul.	Officer	389	1	389	388

Dispatch present means nonblank `TimeDispatch`. Arrival valid means parseable `TimeArrive` no earlier than parseable `TimeCreate`. Every nonblank dispatch and arrival field in these rows is parseable. The table is aggregate and contains no record identifiers.

For record i , define $D_i = 1$ when the eventual released record has a nonblank public dispatch field and $A_i = 1$ when its arrival field parses and is no earlier than call creation. The four states are

$$J_{da,i} = \mathbf{1}\{D_i = d, A_i = a\}, \quad (d, a) \in \{0, 1\}^2. \quad (3)$$

Thus J_{01} means only that the released record contains a valid arrival field and no dispatch field. Calls are binned by creation time; stage fields are read from the eventual record. A later value may have been back-filled or overwritten.

The official metadata calls `TimeDispatch` the time entered when an officer was dispatched, `TimeArrive` the time entered when an officer arrived, and `SelfInitiated` an item generated by an officer in the field rather than one responding to 911 (City of New Orleans, 2021). These are administrative definitions. They do not disclose entry history, overwrite behavior, internal event identifiers, or the transformation from operational systems to the public export.

3.2 First-hand audit of the late-July break

The audit reads the official July 2021 compressed CSV directly. It records field nonblank status, timestamp parseability, valid arrival ordering, state counts, and format classes; no row-level values are retained. All nonblank dispatch and arrival fields in 25–31 July parse under the declared ISO rule, and no sentinel or format change explains the discontinuity.

The audit changes the interpretation in a useful but limited way. It verifies a real public-file discontinuity and shows that initiation streams change together. A complete 14×4 comparison of the seven audited days, two initiation streams, and four states matches the packaged aggregate tally exactly, making an unnoticed aggregate-processing error implausible. Because `SelfInitiated=Y` denotes an item generated by an officer in the field, the disappearance of its public dispatch field while arrival remains complete cannot be read as evidence that the officer was absent. The discrepancy is in the path from activity to released representation, but the public file does not identify whether

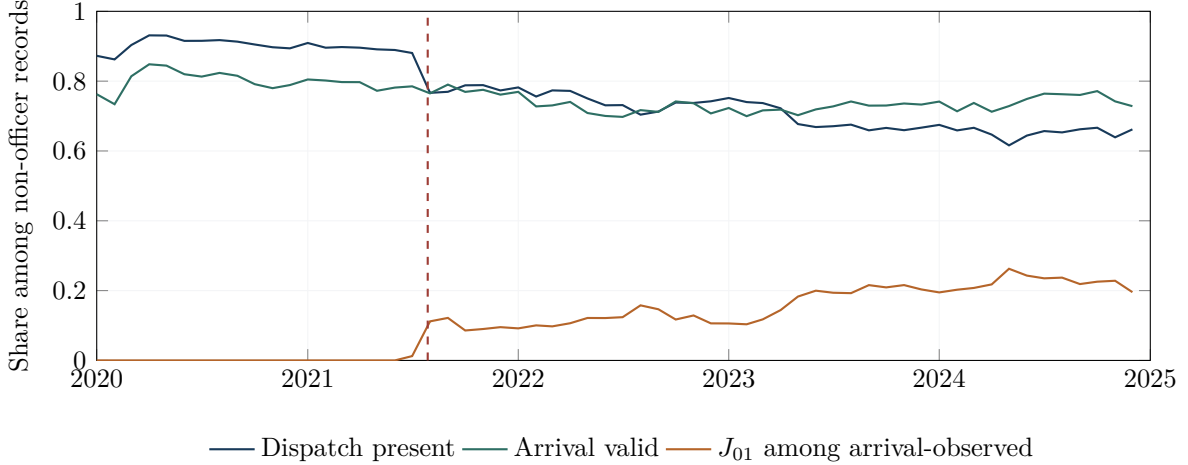


Figure 1. Monthly public-field completeness among non-officer-initiated records. The dashed line marks the first day with nonzero J_{01} . The figure describes released fields, not physical events.

the relevant step was entry, retention, transformation, or export.

4 Comparison design

The Ida event window is 29 August 00:00 through 3 September 00:00, divided into ten twelve-hour call-creation bins. Each bin is compared with the same half-day seven days earlier. The standardized matrix G contains event-minus-baseline contrasts in J_{01} , J_{10} , and J_{11} over common `initialtype` $\times$ `hour-bucket` $\times$ `schema-era` strata, weighted by baseline counts. The headline statistic is

$$M_{\max}(G) = \max_{t,j} |G_{tj}|. \quad (4)$$

The event and its matrix were known before the historical 0.25 sensitivity rule and restricted support library were completed. The three standardized reference universes, their eligibility rules, and context exclusions were fixed before the reference-window outcomes were inspected. The exercise is therefore a transparent descriptive calibration, not a preregistered randomization test.

The principal standardized universe contains 153 ordinary five-day windows starting on or after 1 July 2021. The standardized rule requires at least 0.90 common-support coverage on both event and baseline sides for reference windows. Ida is retained as the target even though it fails that rule, and its coverage is reported rather than concealed.

Two full-count comparisons use all non-officer records in each half-day. The first inherits the prespecified stage-era membership list; it removes common-support selection but remains tied to that universe. The second is support-independent with respect to the July break: both event and baseline sides must begin after 28 July, and prespecified emergency and holiday contexts remain excluded. This post-change universe was defined after the July break was identified and is therefore a post-hoc comparison. It contains 151 references.

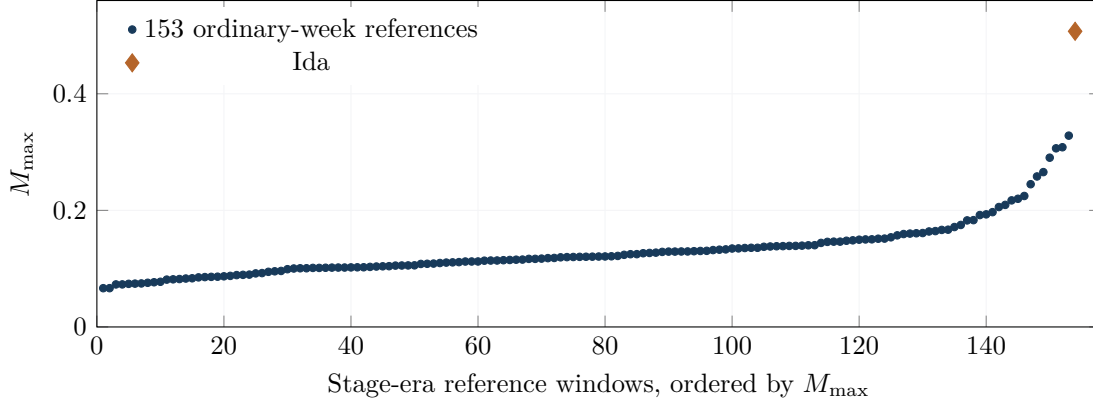


Figure 2. Maximum absolute standardized public-field contrast. The descriptive rank is not a causal or randomized p -value.

5 Results

5.1 Ida is unusually large under both estimators

The standardized Ida statistic is $M_{\max} = 0.5072$, above all 153 stage-era references (rank 1/154 including Ida). The largest reference value is 0.3282. With 4,000 within-stratum multinomial replicates, Ida’s conditional 95-percent interval is $[0.460, 0.566]$; the strongest five reference-window intervals are reported in the supplement.

The unstandardized full-count statistic using all non-officer records is 0.5320. It is rank 1/151 including Ida in the stage-era set and rank 1/152 including Ida in the post-hoc post-change set. Its bootstrap interval is $[0.475, 0.601]$. In a conditional window-wise multinomial bootstrap that independently perturbs Ida and all 151 post-change references without modeling temporal dependence, Ida ranks first in all 4,000 draws. This is a robustness statement about categorical-count perturbations, not formal rank inference for an exchangeable time series.

Alternating the chronological post-change references removes the direct event-to-next-baseline overlap. Ida remains rank 1/77 including Ida in one phase and 1/76 in the other; the largest retained reference values are 0.298 and 0.310. The sensitivity does not create independent random draws, but it shows that the descriptive rank is not an artifact of using both members of each adjacent pair.

The excluded post-Ida week has a full-count contrast of 0.478 because its baseline is Ida itself; it records recovery and is not an independent repeat of the shock. Hurricane Francine, an excluded later-regime hurricane window, is much smaller at 0.110. These labelled comparisons help distinguish Ida’s public-field reconfiguration from a generic later-regime hurricane pattern.

5.2 Standardized support is incomplete

Common-support coverage ranges from 0.748 to 0.928 on the event side and 0.505 to 0.851 on the baseline side. Record-weighted coverage is 0.861 for 3,878 event records and 0.770 for 3,332 baseline records. Ida therefore would not pass the 0.90 rule applied to candidate references. The standardized estimate describes a selected common-support population; the full-count comparison is the primary safeguard against that asymmetry.

The binary exceedance count is also threshold-sensitive. The full-set 95th-percentile maximum-cell threshold is 0.2094 and flags nine cells; the stage-era threshold is 0.2247 and flags eight; the same-season threshold is 0.2057 and flags nine. Continuous contrasts, coverage, and intervals are

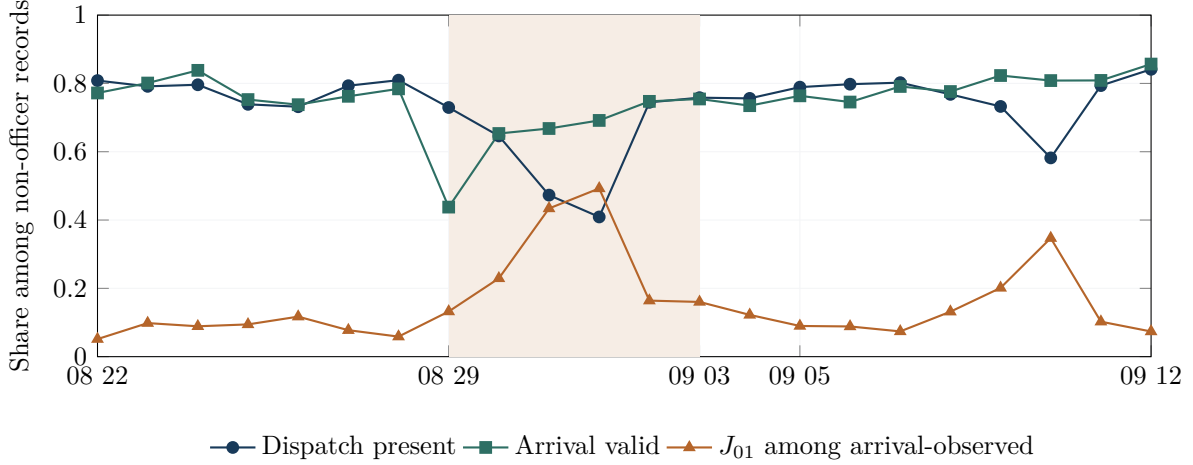


Figure 3. Daily public-field path around Ida. Shading marks the five-day Ida event window. The figure shows the temporary change and recovery in released-field configuration, not physical dispatch or arrival.

Table 2. Prespecified stage-era statistics

Statistic	Ida	Largest reference	Rank including Ida
$M_{\max}$ / amplitude A	0.5072	0.3282	1/154
σ_1	1.0837	0.6011	1/154
σ_2	0.2623	0.2451	1/154
σ_3	0.0858	0.1112	11/154
U_{direct}	0.4223	0.3282	1/154
U_{full} upper endpoint	0.2507	0.2125	1/154
Q upper endpoint	0.4944	0.9408	100/154
D upper endpoint	0.1716	0.2835	4/154

more informative than a single abnormal/normal label.

5.3 Not every statistic is extreme

Ida is first on the simple amplitude, the leading singular values, the direct restricted score, and the constrained LP score, but only 11th on the third singular value, 100th on Q , and fourth on D . The evidence is specifically a large public-field reconfiguration, not uniform extremeness under every summary. The LP remains robustness only: its lower endpoint clears the historical 0.25 rule by 0.000734, and 50 of 217 reference intervals have width above 0.01. A timing exercise is unfavorable because zero-padded shifts fit better than the documented institutional alignment. Neither calculation identifies a mechanism.

5.4 Disposition accounting and initiation streams

Among arrival-observed records, exact Kitagawa decompositions assign the late aggregate dispatch-field contrasts of -0.4950 and -0.4824 primarily to within-recorded-disposition changes (-0.5069 and -0.4946), offset by composition terms of $+0.0119$ and $+0.0123$. Changing observed final-disposition mix is therefore not the sole explanation. Selection into arrival observation, unobserved composition, semantic drift, and recording pathways remain possible.

The initiation-stream pattern is stronger after the raw audit. Officer-initiated dispatch completeness is 99.99 percent in 2020, 61.7 percent over 2021 because the break occurs in late July, and approximately 1 percent in each of 2022–2024. Arrival completeness remains about 99.8–99.9 percent. By contrast, non-officer dispatch completeness declines from 90.3 percent in 2020 to 65.4 percent in 2024. For officer-initiated items, the published definition and nearly complete arrival field rule out interpreting the missing public dispatch field as absence of an officer. The synchronized change across initiation streams therefore documents a coordinated change in released representation. It does not identify whether records were reclassified, different events were retained, or a timestamp was omitted during entry, retention, transformation, or export.

6 What public institutional documents add

The institutional search changes the paper in three ways. First, the DataNOLA page and attached dictionary bind the released labels and provider. The same page warns that records are preliminary, may contain mechanical or human error, may change after investigation, and should not be used for comparison over time; its public changelog reports no archived changes (City of New Orleans, 2021). That warning does not explain the break, but it directly supports the measurement caution.

Second, the public OIPM Hurricane Ida report and emergency materials document the oversight context, outage concerns, curfew, and monitored NOPD activities (Office of the Independent Police Monitor, 2021). Public NOPD policy chapters describe intended communications, call response, and records practices (New Orleans Police Department, 2023). Yet the exact editions in force during July–September 2021 were not located for all relevant chapters, and policy statements do not reveal the historical machine-event to public-export transform. These documents are semantic and institutional context, not a bridge receipt.

Third, two candidate technology explanations are inconsistent with the public chronology. OPCD’s publicly indexed Carbyne APEX cutover is dated 24 June 2022, eleven months after the July 2021 discontinuity (Orleans Parish Communication District, 2022). The New Orleans Inspector General describes Hexagon OnCall Records as a records-management project that was contracted in 2020 but never launched, not as the public CAD export transition (New Orleans Office of Inspector General, 2024). Eliminating these candidates is useful negative evidence; it does not identify the remaining cause.

No agency confirmation of the July 2021 change was available for this version; the City, OPCD, and NOPD were not contacted as part of this public-source revision.

Institutional-evidence boundary

The public documents fill missing semantic, provenance, and exclusion evidence. They do not supply a versioned 2021 changelog, internal dispatch-event history, user-entry audit trail, retention rule, or public-export lineage. Mechanism identification therefore remains open.

7 Current data and denominator discipline

The longer series must compare like with like. In 2024, the non-officer dispatch-field share is 65.38 percent. In the full 2025 file, 136,712 of 207,050 non-officer records have dispatch fields (66.03 percent), compared with 1,117 of 122,720 officer-initiated records (0.91 percent). Pooling both streams gives 137,829 of 329,770 records, or 41.80 percent. The 2026 snapshot is similar: 66.84

Table 3. Current dispatch-field completeness by denominator

Year	Denominator	Records	Dispatch present	Share
2024	Non-officer	212,314	138,804	65.38%
2025	Non-officer	207,050	136,712	66.03%
2025	Officer	122,720	1,117	0.91%
2025	All public rows	329,770	137,829	41.80%
2026 snapshot	Non-officer	130,346	87,129	66.84%
2026 snapshot	Officer	79,483	738	0.93%
2026 snapshot	All public rows	209,829	87,867	41.88%

percent among 130,346 non-officer records, 0.93 percent among 79,483 officer records, and 41.88 percent overall.

The comparable sequence is therefore 65.4 percent in 2024, 66.0 percent in 2025, and 66.8 percent in the 2026 snapshot. The overall 41.8 percent series answers a different question because it combines streams with sharply different recording conventions.

8 Implications for reported domestic-violence research

Public CAD counts are reports created in an administrative system. They are not a direct measure of underlying domestic-violence incidence, victim need, or successful access to help. A defensible analysis should therefore separate four objects:

1. the declared reported-call denominator;
2. the classification rule, retaining $1/0/U$ when label authority is incomplete;
3. the public start and endpoint definitions with coverage by group and period; and
4. the lineage evidence connecting public fields to the target performance measure.

If n_1 records are definitely in the target class, n_0 definitely outside it, and n_U unresolved, then the class share lies in

$$\left[\frac{n_1}{n_1 + n_0 + n_U}, \frac{n_1 + n_U}{n_1 + n_0 + n_U} \right]. \quad (5)$$

Endpoint missingness similarly calls for full-denominator bounds or an explicit selected-stage estimand. Ida can motivate a stress test of these measurement objects, but it cannot by itself establish that domestic-violence incidence or police performance rose or fell.

9 Limitations and conclusion

The analysis uses eventual public records rather than internal event histories. The July break is date-localized in raw official bytes, but its institutional mechanism is not observed. Reference windows are descriptive and not exchangeable by design. Adjacent windows share a week: one window’s event period becomes the next window’s baseline, so a disturbed week can generate two large contrasts of opposite sign and the effective number of independent comparisons is smaller than the nominal count. The bootstrap perturbs windows separately and does not preserve that cross-window dependence or uncertainty in the regime boundary; the alternating-window ranks are a sensitivity, not a time-series correction. Holidays, day-of-week patterns, heat, weather, COVID-era behavior and mobility, reporting and help-seeking thresholds, redaction and suppression, and

changing call composition can affect comparisons. Public `SelfInitiated=N` is not equivalent to “911 only.”

The central conclusion is constructive. Public CAD is valuable when its observable object is stated precisely. In New Orleans, the joint field-presence configuration documents a measurement-regime change before Ida and a further Ida-era stress. Raw audit, denominator repair, source lineage, bootstrap expansion, and institutional documents make that conclusion more secure. They also sharpen what remains missing: the historical bridge from internal events and entry histories to the released fields. Until that bridge is observed, public timestamps should be reported as administrative-field measures rather than treated automatically as operational ground truth.

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